

Bounded Tingley Factorization (Y2), Steps 1–2: well-defined bounded crystal operators and source combinatorics

Seed 4, Round 2 Layer 4 (Warnaar loop experiment)

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Abstract

We prove that the mod- d colored bracket (Tingley-style) crystal operators on *bounded* chains of profile c are well defined ($e_\kappa f_\kappa = f_\kappa e_\kappa = \text{id}$ where defined), uniformly for all $d \geq 2$, without invoking Tingley's $n \geq 3$ hypothesis or his post-publication erratum (Step 1: **proved**). We then classify a family of sources: the *window chains* $(v_{k_1}, \dots, v_{k_{m-1}}, 0)$, $k_1 \geq \dots \geq k_{m-1} \geq 0$, are sources, and they are counted by partitions into parts $\{d, 2d, \dots, (m-1)d\}$, exactly the source combinatorics required by (Y2) (Step 2a-i: **proved**). Completeness of this list (2a-ii) and uniqueness of the source per connected component (2b) are stated as precise gaps with exhaustive computational verification and partial structural results (a shield-acyclicity lemma; nonadjacent-color commutation as a theorem).

1 Setup

Fix $c = (c_0, c_1, c_2) \in \mathbb{Z}_{\geq 0}^3$, $d = c_0 + c_1 + c_2 \geq 2$, $m \geq 1$. Let

$$S = \{a \in \mathbb{Z}_{\geq 0}^3 : a_i \leq a_{i-1} + c_i \ (i \in \mathbb{Z}/3)\},$$

and let X_m be the set of chains $A = (a^{(1)} \geq \dots \geq a^{(m)})$ with all $a^{(s)} \in S$ (componentwise order; the bottom may be 0). X_m is in weight-preserving bijection with cylindric partitions of profile c with largest part $\leq m$; $\text{wt}(A) = \sum_s |a^{(s)}|$.

Boxes, colors, order. A box (i, j, s) ($i \in \{0, 1, 2\}$, $j \geq 1$, $1 \leq s \leq m$) is *present* in A iff $a_i^{(s)} \geq j$. Set $\text{Off} = (0, c_1, c_1 + c_2)$ and extend cyclically: $\widetilde{\text{Off}}(x+3) = \widetilde{\text{Off}}(x) + d$, $\widetilde{\text{Off}}(0, 1, 2) = (0, c_1, c_1 + c_2)$. Define

$$\text{color}(i, j, s) = (s - j + \text{Off}_i) \bmod d, \quad T(i, j, s) = di + 3(j - \text{Off}_i - s) \in \mathbb{Z}.$$

Bracket word and operators. A box (i, j, s) with $j = a_i^{(s)} + 1$ is *column-addable* iff $s = 1$ or $a_i^{(s-1)} \geq j$; a box with $j = a_i^{(s)}$ is *column-removable* iff $a_i^{(s)} > a_i^{(s+1)}$ (with $a_i^{(m+1)} := 0$). *No cross-column S-condition enters the word.* For a color κ , the word $W_\kappa(A)$ lists column-addable boxes of color κ as “(” and column-removable ones as “)”, sorted by T increasing. Cancel “(” pairs stack-wise; f_κ adds the box of the leftmost surviving “(”; e_κ removes the box of the rightmost surviving “)”; $\varphi_\kappa, \varepsilon_\kappa$ count surviving “(”, “)”.

2 Step 1: the operators are well defined (all $d \geq 2$)

Lemma 1 (No ties). *Two distinct boxes of the same color have distinct T .*

Proof. Suppose $(i, j, s) \neq (i', j', s')$ have equal color and equal T , both being column-addable or column-removable corners. Let $\alpha = (j - \text{Off}_i - s) - (j' - \text{Off}_{i'} - s')$. Equal T gives $d(i - i') = -3\alpha$; equal color gives $d \mid \alpha$. Writing $\alpha = dt$ yields $i - i' = -3t$, and $|i - i'| \leq 2$ forces $t = 0$, hence $i = i'$ and $j - s = j' - s'$: two corners of the same column on the same diagonal. This is impossible: for two addables at levels $s < s'$, $j - s = j' - s'$ gives $a_i^{(s)} - a_i^{(s')} = s - s' < 0$, contradicting monotonicity; similarly for two removables; and an addable at level s and a removable at level s' on one diagonal force $a_i^{(s')} - a_i^{(s)} = s' - s + 1$, which by monotonicity requires $s' = s - 1$ and $a_i^{(s-1)} = a_i^{(s)}$, contradicting addability at level $s \geq 2$ (and $s' = 0$ is not a level). Machine-checked as C0 (`seed4.R2L4_step1.py`): no same-color T -ties over 26371 chains, $d \in \{2, \dots, 9\}$. $\square$

Lemma 2 (Locality). *Adding or removing one box (i, j, s) changes the letter multiset of W_κ only at the box itself (color $\kappa_0 = \text{color}(i, j, s)$) and possibly at the four neighboring corner boxes $(i, j \pm 1, s)$, $(i, j, s \mp 1)$, whose colors are $\kappa_0 \pm 1 \pmod d$. In particular, for $d \geq 2$ and any $\kappa \notin \{\kappa_0 - 1, \kappa_0, \kappa_0 + 1\}$, W_κ is unchanged, and for $\kappa = \kappa_0$ the only change is the flip of the letter at $T(i, j, s)$.*

Proof. Column-addability/removability of a box (i', j', s') depends only on the three column values $a_{i'}^{(s'-1)}, a_{i'}^{(s')}, a_{i'}^{(s'+1)}$. Changing $a_i^{(s)}$ by ± 1 affects only boxes with $i' = i$ and $s' \in \{s - 1, s, s + 1\}$, and there only corners at $j' \in \{j - 1, j, j + 1\}$ can change status. Their colors differ from κ_0 by $(s' - s) - (j' - j) \in \{0, \pm 1\}$, and the value 0 occurs only for the box itself or the diagonal neighbors $(i, j \pm 1, s \pm 1)$, whose addability/removability status is unaffected by the change at (i, j, s) (their defining column triples do change only in coordinates that do not affect their corner status; direct check of the finitely many cases). Machine-checked as CL (same script): 146784 operations, zero exceptions. $\square$

Lemma 3 (Validity witness). *Let (i, j, s) be column-addable in $A \in X_m$ but suppose $A + (i, j, s) \notin X_m$ (the S -condition $a_i \leq a_{i-1} + c_i$ fails at level s). Then $(i - 1, a_{i-1}^{(s)} + 1, s)$ is column-addable, has the same color, and sits at $T - d$; moreover no color- κ_0 letter lies strictly between $T - d$ and T . Hence the invalid “(” is never the leftmost surviving “(” (the witness “(” either survives to its left or cancels against a “)” strictly left of T , forcing a surviving “(” left of T in either case by the stack rule). Dually for removals with witness at $T + d$.*

Proof. Failure of the S -condition after adding means $a_i^{(s)} + 1 > a_{i-1}^{(s)} + c_i$, i.e. $a_{i-1}^{(s)} \leq a_i^{(s)} - c_i$; combined with $a_{i-1}^{(s)} \geq a_i^{(s)} - c_i$ (from $a^{(s)} \in S$) we get $a_{i-1}^{(s)} = a_i^{(s)} - c_i =: j' - 1$ with $j' = j - c_i$. Column-addability of $(i - 1, j', s)$: for $s \geq 2$, $a_{i-1}^{(s-1)} \geq a_i^{(s-1)} - c_i \geq j - c_i = j'$ using $a^{(s-1)} \in S$ and $a_i^{(s-1)} \geq j$ (addability of (i, j, s)); for $s = 1$ automatic. Color: $s - j' + \text{Off}_{i-1} = s - j + c_i + \text{Off}_{i-1} \equiv s - j + \text{Off}_i \pmod d$. Position: $T(i - 1, j', s) = T(i, j, s) - d$ (direct computation, also for $i = 0$ using $\widetilde{\text{Off}}$). The gap $(T - d, T)$ contains no color- κ_0 boxes because color- κ_0 boxes occupy an arithmetic progression of step d in T (as $T \equiv di - 3(\text{Off}_i + \kappa_0) \pmod{3d}$ on color κ_0 , consecutive slots differ by d). Machine-checked as CV: 35292 S -violating boxes encountered, witness present and correctly placed in all. $\square$

Lemma 4 (Signature flip). *Let w be a bracket word and let p be the position of the leftmost surviving “(”. Flipping p to “)” makes p the rightmost surviving “)” of the new word, and the surviving multiset changes by exactly $(\varphi, \varepsilon) \mapsto (\varphi - 1, \varepsilon)$ at that color with the new “)” surviving. Dually for the rightmost surviving “)”.*

Proof. Stack-depth argument. The prefix before p is unchanged and contains no surviving “(”. In the new word the stack at p is as in w ; the new “)” at p survives iff the stack is empty there, which

holds because p was the *leftmost* surviving “(” of w (any unmatched “(” before p would survive in w). To the right of p , matchings are unchanged: if some “)” at $q > p$ first emptied the new stack, then in w that “)” would have popped p — contradicting that p survives in w . Hence the new surviving “)”-set is the old one plus p , and p is rightmost among them because all old surviving “)” lie left of all old surviving “(”.

Theorem 5 (Step 1). *For all $d \geq 2$ and all profiles c : f_κ and e_κ map X_m to $X_m \cup \{0\}$, and $e_\kappa f_\kappa = \text{id}$ on $\{f_\kappa \neq 0\}$, $f_\kappa e_\kappa = \text{id}$ on $\{e_\kappa \neq 0\}$. Moreover $\varphi_\kappa(f_\kappa A) = \varphi_\kappa(A) - 1$, $\varepsilon_\kappa(f_\kappa A) = \varepsilon_\kappa(A) + 1$.*

Proof. Closure: by Lemma 3 the leftmost surviving “(” is never S -invalid, and column-validity ($a_i^{(s-1)} \geq j$, $a_i^{(s)} \geq a_i^{(s+1)}$ after the change) is built into column-addability/removability; so $f_\kappa A \in X_m$ when defined, dually for e_κ . Inverse: by Lemma 2 the color- κ word of $f_\kappa A$ differs from that of A only by the flip at the added box; by Lemma 4 that box is the rightmost surviving “)” of $W_\kappa(f_\kappa A)$, so e_κ removes exactly it. Ties never obstruct (Lemma 1). Machine verification: 0 failures over 26371 chains and 146784 operator applications, $d \in \{2, \dots, 9\}$ including $3 \mid d$ (`seed4_R2L4_step1.py`, checks C0/CL/CV/CS/CE). $\square$

Remark 6. The proof nowhere uses Tingley’s hypothesis $n \geq 3$ nor the erratum-affected part of his §4.2: the bracket word is defined *column-locally*, so the only interaction between columns is the witness of Lemma 3, which is an immediate neighbor at $T \mp d$. In particular $d = 2$ is covered uniformly, resolving adjudication (vi) of the Layer-3 synthesis.

3 Step 2a: window chains are sources

Definition 7 (Window vectors and v -chains). For $k \geq 0$ let $v_{k,i} = \sum_{t=0}^{k-1} c_{(i-t) \bmod 3} = \widetilde{\text{Off}}(i) - \widetilde{\text{Off}}(i - k)$. Then $|v_k| = kd$, $v_k \in S$, $v_k \leq v_{k+1}$. For $k_1 \geq \dots \geq k_{m-1} \geq 0$ the v -chain is $A = (v_{k_1}, \dots, v_{k_{m-1}}, 0) \in X_m$.

Theorem 8 (2a-i). *Every v -chain is a source: $\varepsilon_\kappa(A) = 0$ for all κ .*

Proof (per-boundary matching). Set $k_m := 0$. For a boundary $1 \leq s \leq m - 1$ let $\delta = k_s - k_{s+1}$, $x = -k_s$, and window $w_i = \widetilde{\text{Off}}(x + i + \delta) - \widetilde{\text{Off}}(x + i) \geq 0$. Direct computation from the definitions gives: the letters of the full word attributable to boundary s are, per slot $j \in \{0, 1, 2\}$, with $F(y) = d(y - x) - 3\widetilde{\text{Off}}(y) - 3s$ (note $F(y + 3) = F(y)$):

- “)” at $T = F(x + j)$, color $(s + \widetilde{\text{Off}}(x + j)) \bmod d$, present iff $w_j > 0$;
- “(” at $T = F(x + j) - d\delta$, *same color*, present iff $w_{(j-\delta) \bmod 3} > 0$.

(The level-1 addables form boundary 0 and are extra “(”s; there are no removables at level m since $a^{(m)} = 0$.) Since boundaries partition the letters, it suffices to exhibit, *within each boundary and each color*, an injection from “)”s to strictly earlier “(”s; the union over boundaries is injective and implies the ballot condition, hence $\varepsilon_\kappa = 0$.

Count equality. At each boundary and color, $\#“)” = \#“($: the multisets $\{\widetilde{\text{Off}}(x + j) \bmod d\}_j$ and $\{\widetilde{\text{Off}}(x + i + \delta) \bmod d\}_i$ coincide (three consecutive arguments of $\widetilde{\text{Off}}$), and a column with $w_i = 0$ deletes the same value from both.

Matching. (M1) If $\delta \geq 3$: every $w_i \geq \lfloor \delta/3 \rfloor d > 0$, all six letters exist, and the slot-wise pairing “(” at $F_j - d\delta < F_j$ works. (M2) If $\delta \equiv 0 \pmod{3}$: the partner column $(j - \delta) \bmod 3 = j$, so existence is equivalent on both sides and the slot-wise pairing works. (M3) If $\delta \in \{1, 2\}$: a partner may be

missing, but count equality supplies a same-color replacement “(” in another slot j' ; the required inequality $F_{j'} - d\delta < F_j$ involves only $F_{j'} - F_j = d(j' - j) - 3\Delta$ with $\Delta = \widetilde{\text{Off}}(x + j') - \widetilde{\text{Off}}(x + j) \in \{0, \pm d\}$ on a common color class, so every comparison is a multiple of d determined by $(j, j', \text{sgn } \Delta, \delta)$ and the existence flags, which are determined by the zero pattern of c and $x \bmod 3$. This finite space was checked exhaustively: all compositions c of $d \leq 12$ (every zero pattern and value coincidence), generic large profiles per zero pattern, $x \bmod 3 \in \{0, 1, 2\}$, $\delta \in \{1, \dots, 9\}$: 25272 boundary/color instances, ballot holds in all, no T -ties (`seed4_R2L4_boundary.py`). Invariance under $x \mapsto x + 3$ (F periodic, colors shift by $d \equiv 0$, w unchanged) and (M1) for $\delta \geq 3$ make the sweep exhaustive for the reduced data. $\square$

Corollary 9 (Source generating function, lower bound). *The map $(k_1 \geq \dots \geq k_{m-1}) \mapsto \text{conjugate partition}$ realizes a weight-preserving bijection between v -chains and partitions into parts $\{d, 2d, \dots, (m-1)d\}$; $\text{wt}(A) = d \sum_s k_s$. Hence*

$$\sum_{v\text{-chains}} q^{\text{wt}} = \prod_{j=1}^{m-1} \frac{1}{1 - q^{jd}},$$

exactly the source combinatorics required by (Y2).

4 Step 2a-ii and 2b: precise gaps and partial results

Lemma 10 (Shield acyclicity). *Call a removable r shielded by an addable p if they have the same color and $T(p) < T(r)$. $\varepsilon_\kappa(A) = 0$ for all κ iff there is a globally injective assignment of shields. Any such assignment strictly decreases T along shield chains, so its digraph is acyclic and every chain terminates at an addable that is not the T -successor of a removable (“free” addable).*

Proof. The injection criterion for the ballot condition is standard (Hall-type: each prefix contains the distinct shields of its “)”s). Acyclicity is immediate from T -decrease. $\square$

Conjecture 11 (2a-ii, completeness). *Every source of X_m is a v -chain.*

Evidence and partial proof. (i) Exhaustive verification: sources coincide with v -chains in 16 cases (c, m, W) , $d \in \{2, \dots, 8\}$, $m \leq 4$ (`seed4_R2L4_step2.py`, `checkA`). (ii) The base case $m = 1$ is verified for $d \leq 10$, all profiles c , weights $\leq 3d + 6$: the only source is 0. (iii) By Lemma 10, a nonzero source would carry shield chains ending at free addables; at $m = 1$ the color/position congruences ($u_{i'} \equiv u_i - 1 \pmod d$ with $u_i = a_i - \text{Off}_i$, plus T -inequalities) contradict the S -inequalities $u_0 \geq u_1 \geq u_2 \geq u_0 - d$ in the endpoint cases worked so far; the remaining endpoint cases and the induction on m are open.

Theorem 12 (Nonadjacent commutation). *If $\kappa \not\equiv \lambda, \lambda \pm 1 \pmod d$ then $e_\kappa e_\lambda = e_\lambda e_\kappa$ and likewise for f ’s, wherever defined.*

Proof. By Lemma 2, applying e_λ leaves W_κ untouched, and conversely; both compositions remove the same pair of boxes. $\square$

Conjecture 13 (2b, unique source per component). *Every connected component of $(X_m, e_\kappa, f_\kappa)$ contains exactly one source.*

Evidence and obstruction. Exhaustive union-find over e -edges: every W -truncated component has exactly one source, 0 violations in 6 cases $d \in \{2, 3, 4, 5, 7\}$ (checkC); moreover *global confluence* (the set of sources reachable by arbitrary e -descents from any chain is a singleton) verified in 9 cases, up to 2968 chains per case (`seed4_R2L4_step2b.py`, check D). Obstruction to the natural proof: adjacent-color one-step divergences have unbounded meet depth (observed depths up to 9 at $W = 10$; check F), so no fixed-depth local-confluence (Stembridge-type) lemma exists and Newman’s lemma with bounded search is unavailable. Suggested routes: an explicit crystal morphism onto the vacuum component, or a string-length invariant.

5 Verification summary

All scripts in 2026-07-04/scratch/scripts/: `seed4_R2L4_step1.py` (Step 1: C0/CL/CV/CS/CE; 26371 chains, 146784 ops, 35292 S -violating boxes, all pass); `seed4_R2L4_sources.py`, `seed4_R2L4_step2.py` (source classification, GF match, union-find); `seed4_R2L4_match.py`, `seed4_R2L4_step2b.py` (matching recon; global confluence; meet-depth histogram); `seed4_R2L4_boundary.py` (the (M3) finite case check, 25272 instances, 0 failures; plus $\varepsilon = 0$ re-verified on 684 v -chain cases, 0 surviving removables).